

Executive Summary

HealthConnect Week 7 Executive Summary

Week 7 focused on validating the analytical findings and KPIs developed during Week 6 using independent Excel calculations and pivot-table analysis. The validation examined overall appointment outcomes, previous no-show behaviour, reminder-channel performance, distance to clinic, booking lead time and appointment type.

The dataset contains **5,000 appointments**, of which **2,314 were attended, 2,423 were no-shows and 263 were cancelled**. This corresponds to an attendance rate of **46.28%**, a no-show rate of **48.46%**, and a cancellation rate of **5.26%**.

The validation identified an important KPI labelling/calculation issue in the Excel validation sheet: the cell labelled "**No-Show Rate**" **currently displays 5.26%**, which corresponds to the cancellation rate rather than the no-show rate. Based on 2,423 no-shows out of 5,000 appointments, the correct no-show rate is **48.46%**. This should be corrected before final reporting.

The validation further supported the Week 6 finding that previous no-show behaviour is associated with higher current no-show rates. Patients with no previous no-show recorded a **43.51%** no-show rate, compared with **53.49%** for one previous no-show. The rate increased further for patients with multiple previous no-shows, reaching **59.36%, 67.95%, 66.67%** and **100%** for previous-no-show counts of 2–5 respectively. However, the groups with 4–5 previous no-shows should be interpreted cautiously because their sample sizes may be very small.

Distance to clinic also remained an important finding. The overall no-show rate increased across several distance categories, with patients living **above 30 km** recording the highest observed rate at **68.06%**.

Reminder-channel analysis showed different observed no-show rates: **SMS 45.75%, Email 48.41%, WhatsApp 49.77%, and No Reminder 51.39%**. This supports further investigation of reminder strategies, but it does not establish that a particular channel caused the lower or higher no-show rate.

Booking lead time showed a stronger pattern. The overall no-show rate was **27.81% for 0–7 days, 33.55% for 8–14 days, 43.21% for 15–30 days, and 60.49% for 31–60 days**. This indicates that longer booking lead time deserves particular attention in subsequent modelling and decision support.

Appointment-type validation showed no-show rates of **51.23% for Follow-up, 47.44% for Specialist Consultation and 46.64% for General Consultation**, maintaining the Week 6 observation that follow-up appointments had the highest observed no-show rate.

Overall, Week 7 strengthened the evidence supporting behavioural, accessibility and scheduling-related factors as important areas for no-show analysis. The results should now be provided to the Data Science track for predictive validation before finalising the risk-segmentation and intervention strategy.